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PART 1

(a) No, sometimes softwares do not have a strong error-handling or self-check evaluation which increases chances of failure which are dangerous in the practical environment. In order to have reliable software, the author suggests that worst-case scenarios be created and design the software to respond accordingly. Most of the time engineers have the assumption that the average case scenario will occur all the time, and therefore the software does not have a strong defensive protocol to follow.
(b) Safety should be the first thing as the author describes to design the worst case scenario and construct the software in a way that it responds to such cases. The approach that should be taken is listing out all possible errors the software can face and write down how the software should respond to each. All of this should happen before the implementation of the software.
(c) It is safer to build from scratch. The author explains that just because a software is safe in one system, it does not mean that it is safe in another system.
(d) This is not necessarily true, as it depends what the goal of the software is.The author explains that object oriented design is useful when handling data but it is not a good solution when approaching control-oriented systems.
(e) It is better to first implement error handling and then normal behaviour. This is because when you have an idea of the software you are going to design it is better to list out all possible errors the software can face so you can design how to counter the problems. When engineers think about errors after designing the program, it can cause them to miss an error as they may only be thinking within the boundaries of their design.

PART 2

Develop a use case model (use cases and use case diagram that relates these use cases) for the process of installing an elevator as presented in the video.

USE CASE 1: Creating foundation of elevator after shaft is prepared

Primary Actor(s): Technicians, elevator construction workers

Stakeholders and Interests:
Technicians - who help to accurately align the brackets of the elevator
Construction Workers - who take measurements to install parts of the elevator

Pre-condition(s):
Elevator shaft is created with proper measurements to begin installation of the elevator

Success Guarantee(s): Elevator rails and rail brackets are installed perfectly on each floor, hydraulic pistons are working properly to ensure upward and downward motion of the elevator. Sensors are monitoring to ensure pistons are moving in sync, a computerized motion control system is installed and control jump is installed to allow installation of a temporary run box.

Main Success Scenario:

1. Spot brackets are placed at the top most part of the elevator shaft
2. Lower brackets are aligned with spot brackets with a plumb line
3. Rail brackets are perfectly straight and aligned within 1/64th of an inch
4. Rail brackets are placed on each floor and on each side
5. Guide rails are attached with assistance of 1 ton chain hoist
6. Hydraulic pistons are operating perfectly, upward motion is achieved with oil being pumped into pistons, elevator lowers when pressure is released in system
7. Sensors are installed to ensure pistons are working in sync on both sides of elevator
8. Selector is installed to ensure precise alignment of elevator with floor
9. Computerized motion control system is installed
10. Control jump is installed to allow installation of temporary run box

Extensions:

2a. Lower brackets are not perfectly aligned with spot brackets

    2a1. Make sure weight at end of plumb line is 12 lb to help reduce sway and ensure precise measurement

6a. Elevator is not moving upwards properly

    6a1. Oil pump needs to be replaced in order to allow oil to be pumped into the hydraulic pistons

    6a2. Hydraulic pistons need to be replaced with new ones in order to provide energy to push elevator upwards

6b. Elevator is not moving downwards properly

    6b1. Elevator pistons need to be replaced as pressure is not being released to lower the elevator

7a. Pistons are not working in sync causing elevator to not move smoothly up and down the floors

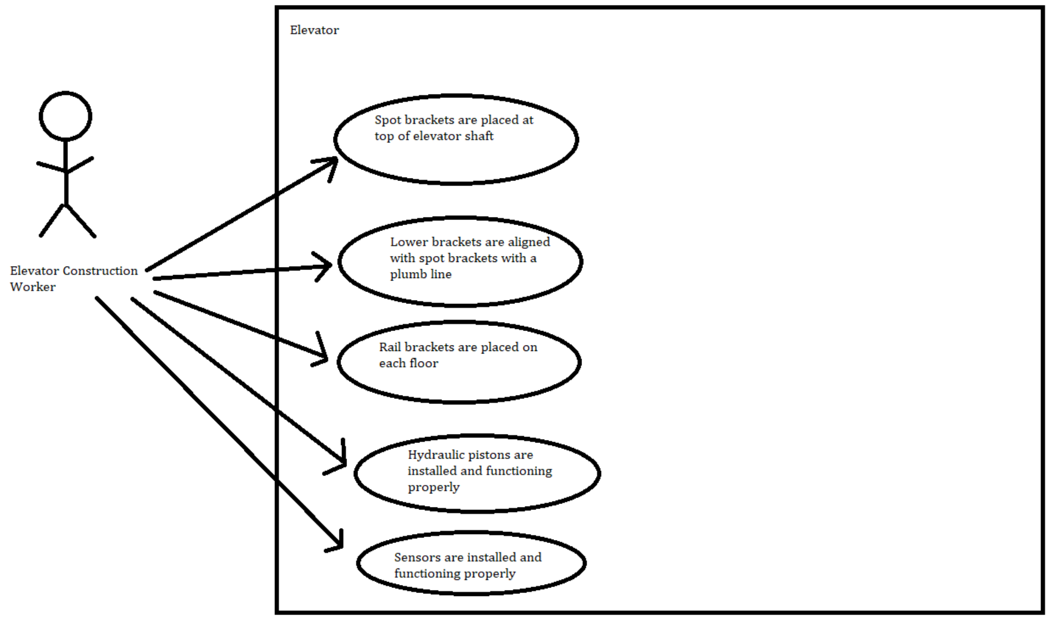
    7a.1 Sensors need to be checked for functionality and if required replaced with new ones to ensure the pistons are running in sync.

8a. Elevator is not stopping in correct alignment to the floor

    8a1. The position magnets on selected tape need to be repositioned to ensure alignment

10a. Temporary run box is not causing the elevator to move up and down.

    10a1. Control jump needs to be installed correctly to make sure the run box is connected to the elevator.



USE CASE 2: Cab assembly

Primary Actor(s): Technicians, elevator construction workers

Stakeholders and Interests:
Technicians - who help to accurately align the brackets of the elevator
Construction Workers - who take measurements to install parts of the elevator
Steelworkers - who place the sills around the elevator
Fire Service - who inspect the elevator for emergency fires
Electricians - who make sure wires are connected properly for elevator

Pre-condition(s):
Elevator shaft is ready. Elevator foundation is prepared with brackets, pistons, sensors and a temporary run box.

Success Guarantee(s):
Elevator cab is assembled successfully allowing it to move upward and downwards without any collisions.

Main Success Scenario:

1. Precise measurements are taken to install struts
2. Brackets are installed to hold struts in place
3. Hoist sill is placed at the entrance on each level
4. Marks are made for door installation

5. Header is placed along the top of the entrance at each landing completing the frame of the door which can now be installed
6. Landing door is going to be installed
7. Side and interior walls sit on top of the platform to begin assembly of cab
8. Strike column and return column are installed and the front panels of the car are assembled for installation of dome
9. Team unpacks the dome and ceiling unit to finish off the top of the elevator
10. Door control and motor drive unit resides atop the front of the cab and is joined to the front walls and the dome to structurally tie all panels together
11. Cab door is moved into place and attached to the door operator and adjustments are made until the door is running smoothly

Extensions:

*a.Measurements do not have a good reference point which is causing mis-alignment of doors and entrances

  *a. Struts are not placed properly as they are used to guide the installation at each landing

2a. Elevators are in danger of collision and are not moving at a proper distance of each other

  2a1. Brackets are to be installed in precise locations to hold the struts securely in place

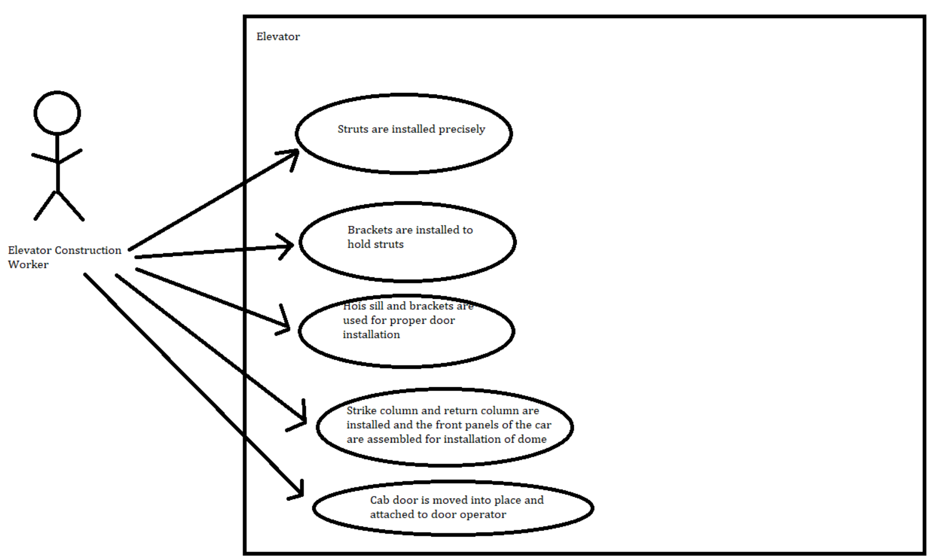
3a. Doors are not opening and closing smoothly

  3a1. The hoistway sill and cap sill are not perfectly aligned and need to be repositioned for smooth opening and closing of the door.

  3a2. The header needs to be repositioned in order to allow smooth opening and closing

8a. Commands are not being received successfully from the elevator

  8a1. The cars' vital electronic command and control assembly are not installed properly from the strike column and return column.



USE CASE 3: Door installation on the elevator

Primary Actor(s): Technicians, elevator construction workers

Stakeholders and Interests:
Technicians - who help to accurately align the brackets of the elevator
Construction Workers - who take measurements to install parts of the elevator
Steelworkers - who place the sills around the elevator
Fire Service - who inspect the elevator for emergency fires
Electricians - who make sure wires are connected properly for elevator

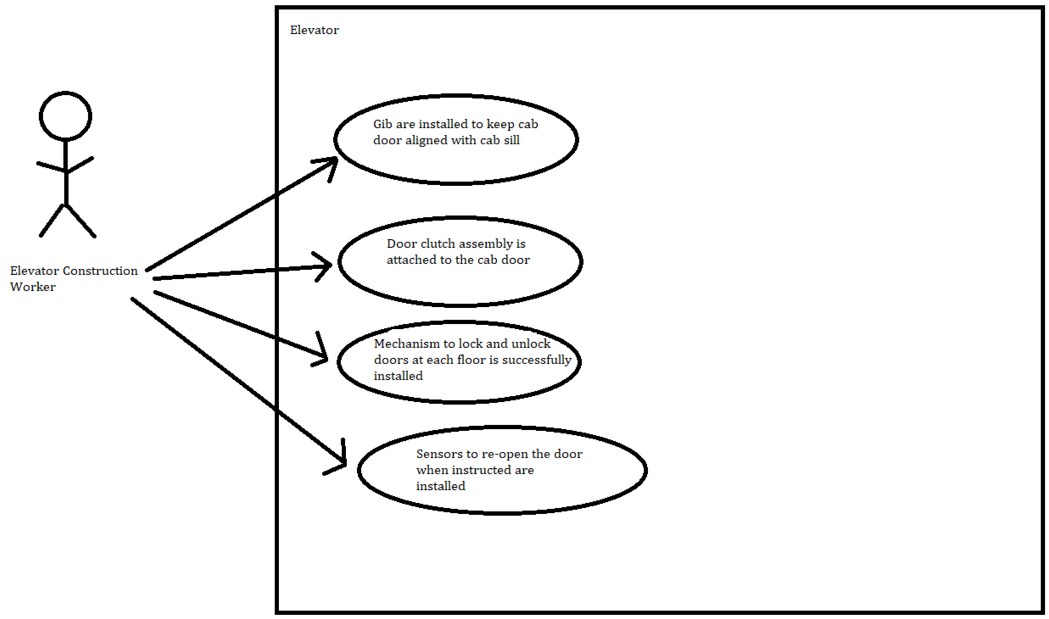
Pre-condition(s):
Cab has been successfully installed and is ready for door installation.

Success Guarantee(s):
Door is installed successfully and operates without any interruptions.

Main Success Scenario:

1. Gib are installed to keep cab door aligned with cab sill
2. Door clutch assembly is attached to the cab door
3. Mechanism to lock and unlock doors at each floor is successfully installed
4. Sensors to re-open the door when instructed are installed

Extensions:
1a. Cab door is not opening properly
    1a1. Gibs are to be placed with precise measurements so the door is aligned with the cab sill.
3a. Door is sliding open and closed while elevator is moving
    3a1. The lock mechanism is to be replaced with a new one to ensure the elevator doors are locked properly.
4a. Door is not opening when the elevator is reaching a certain floor.
    4a1. The sensors have are to be tested for functionality and placed in correct location to ensure the doors open successfully.
    4a2. Door safety sensors are checked to make sure they open if the light curtain is interrupted, i.e. if someone puts their hand in front of the door it causes the door to open again.



USE CASE 4: Electric installation with wiring to allow control systems to function properly

Primary Actor(s): Technicians, elevator construction workers

Stakeholders and Interests:
Technicians - who help to accurately align the brackets of the elevator
Construction Workers - who take measurements to install parts of the elevator
Steelworkers - who place the sills around the elevator
Fire Service - who inspect the elevator for emergency fires
Electricians - who make sure wires are connected properly for elevator

Pre-condition(s):
Elevator door is successfully installed and ready for electrical working.

Success Guarantee(s):
Electrical wiring and connections are made successfully and the elevator can now function with computer and control systems.

Main Success Scenario:

1. Crew install wires to the computer and control systems to allow elevator functionality

2. Jumpers that override the computer control system are removed
3. Temporary run box to move the elevator is now removed
4. Speed of elevator is fine-tuned to appropriate speed
5. Computer is programmed with various command and control protocols required for the proper operations, the elevator is turned over to full automatic control.

Extensions:

*a. Elevator is not responding appropriately as per wire connections

*a1. High voltage wires are to be used for electrical systems, and low voltage wiring is to be run between computer and various devices and sensors

*a2. Wiring of computer and control systems are connected with a cable with hundreds of wires, so ensure the wires are not tangled and successfully travel through the cable.

*a3. The crew must double check that all car control systems are wired correctly.

*a4.Wires fixed to the non-moving systems are carried through solid conduit (metal pipe), wires that move with the car are carried via travelling cable.

3a. Elevator is not moving without the temporary run box.

3a1. Inspection station on top of the elevator is not installed properly on top of the car, the station needs to be tested before proceeding with installation.

4a. Elevator speed is not appropriate

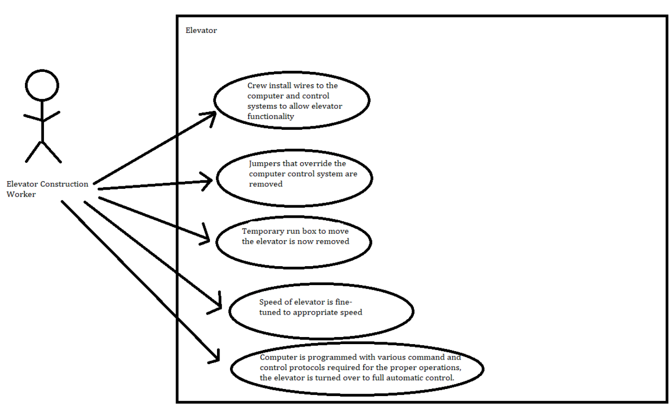
4a1. Team member needs to keep track of elevator speed properly while other team a member controls the elevator from the control room.

4a2. Valves are to be adjusted properly to allow proper rate at which hydraulic fluid is passed into the pistons.

5a. Elevator is not responding to commands given by buttons

5a1. Elevator buttons, switches and systems are to be tested to make sure elevator is functioning properly

5a2. Fire systems are checked to make sure that the elevator recalls to the proper floors in case of an emergency.



PART 3

Primary Actor(s): Passenger

Stakeholders and Interests:
Help Service- who help to accurately align the brackets of the elevator
Fire Service - who inspect the elevator for emergency fires
Electricians - who make sure wires are connected properly for elevator

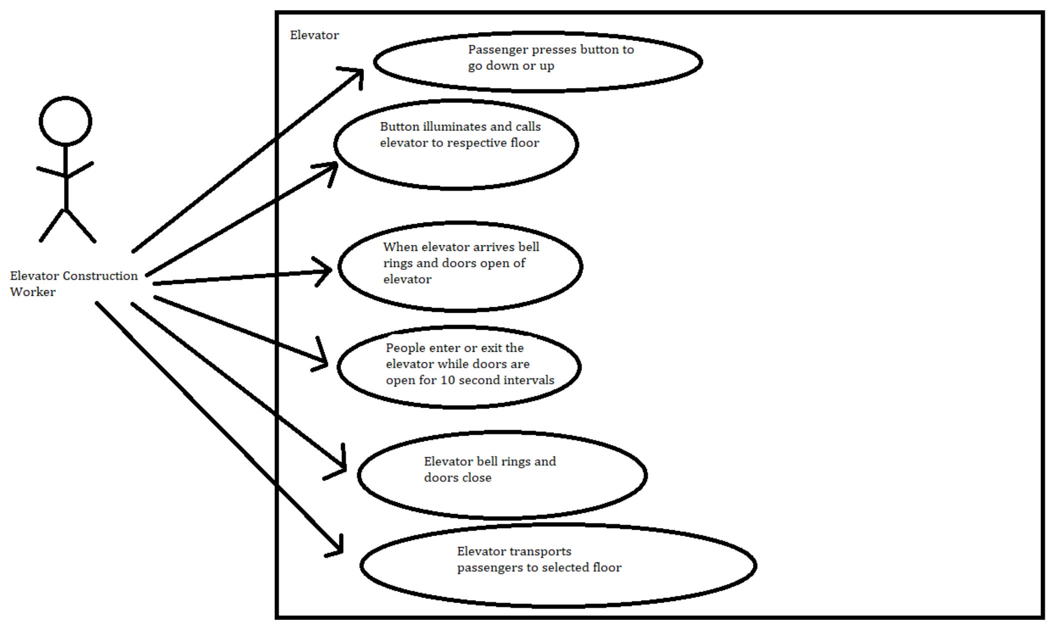
Pre-condition(s):
Elevator is successfully installed and operating properly.

Success Guarantee(s):
Passenger enters elevator, selects floor, elevator takes passenger to selected floor and passenger exits elevator.

Main Success Scenario:

1. Passenger presses button to go down or up
2. Button illuminates and calls elevator to respective floor
3. When elevator arrives bell rings and doors open of elevator
4. People enter or exit the elevator while doors are open for 10 second intervals
5. Elevator bell rings and doors close
6. Elevator transports passengers to selected floor

Extensions:
4a. Too many people enter the elevator causing weight to go over the capacity
    4a1. Elevator does not move and an audio and text message is presented asking passengers to exit the elevator.
4b. There are passengers entering while the elevator door is closing
    4b1. The elevator door has a sensor that opens the door again if the door closing has been interrupted. If too many passengers still proceed a warning will be sent via audio and text message.
6a. The elevator has stopped and there is no power outage or any such emergency
    6a1. The help button contacts 911 if there is no response within 5 seconds of pressing the button
6b. There is a fire while people are travelling in the elevator
    6b1. The elevator will travel to a safe floor and allow passengers to exit safely. There is also a fire alarm signal button that takes the elevator to a safe floor. An audio and text message will be sent to the passengers informing them of the emergency.
6c. There is a power outage while passengers are travelling in the elevator.
    6c1. In the event of a power outage the elevator will use reserved power and transport the passengers to a safe floor to exit safely. The elevator will also notify the passengers of the power outage via audio and text message.


Elevator
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Elevator Construction Worker